

Hallucination Detection Report

Generated: 2026-04-14 09:16 · Extracted triples: 7 · Verdicts: 4 · Inconsistencies: 0 · Symbolic violations: 3

Executive Summary

2 Correct	1 Predicate Mismatch	1 Fabricated Fact
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Extracted Triples

All verifiable triples extracted from the LLM response before any canonicalization or checking.

Subject	Predicate	Object	Verifiable	Verification Type	Predicate Definition
575	HAS_INTEREST_RATE	12.21%	Yes	ENTITY_ATTRIBUTE	The interest rate directly associated with the subject (e.g., a FICO score).
580	HAS_INTEREST_RATE	11.91%	Yes	ENTITY_ATTRIBUTE	The interest rate directly associated with the subject (e.g., a FICO score).
579	HAS_ESTIMATED_INTEREST_RATE	~12.06% (estimated)	Yes	FACTUAL_CLAIM	The final estimated interest rate asserted for the subject, derived from logical extrapolation.
579	HAS_RISK_TIER	Subprime	Yes	ENTITY_ATTRIBUTE	The risk tier category assigned to the subject (e.g., a FICO score).
890	HAS_ESTIMATED_INTEREST_RATE	~4.36% or lower	Yes	FACTUAL_CLAIM	The final estimated interest rate asserted for the subject, derived from logical extrapolation beyond the provided range.
890	HAS_RISK_TIER	Super Prime	Yes	ENTITY_ATTRIBUTE	The risk tier category assigned to the subject (e.g., a FICO score).
835	HAS_ESTIMATED_DOWN_PAYMENT	~2.8% or lower	Yes	FACTUAL_CLAIM	The final estimated down payment asserted for the subject, derived from logical extrapolation.

Canonicalized Triples

Triples after predicate normalization against the database schema. Shows the original predicate, the matched canonical predicate, and match confidence.

Subject	Original Pred.	Object	Canonical Pred.	Confidence
575	HAS_INTEREST_RATE	12.21%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
580	HAS_INTEREST_RATE	11.91%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
579	HAS_ESTIMATED_INTEREST_RATE	~12.06% (estimated)	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
579	HAS_RISK_TIER	Subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
890	HAS_ESTIMATED_INTEREST_RATE	~4.36% or lower	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
890	HAS_RISK_TIER	Super Prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
835	HAS_ESTIMATED_DOWN_PAYMENT	~2.8% or lower	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	MEDIUM

Logical Inconsistency Report

Contradictions detected between triples in the LLM response — independent of the database. These are self-contradictions within the answer itself.

No logical inconsistencies detected.

Symbolic Rule Violations

Violations caught by explicit domain constraints before database lookup. Violated triples were removed from the evidence retrieval step.

Constraint ID	Rule Type	Severity	Subject	Predicate	Object	Explanation
c_tier_300_579		MEDIUM	579	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	Subprime	Credit scores the subject 300-579 must be classified as 'deep subprime' the object. The object 'Subprime' does not match the required classification.
c1_credit_score_range		MEDIUM	890	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	~4.36% or lower	Credit score the Subject must be between 300 and 850 inclusive. The subject '890' is outside the valid range.
c_credit_score_range		MEDIUM	890	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	Super Prime	Credit score the Subject must be between 300 and 850 inclusive. The subject '890' is outside the valid range.

Detected Hallucinations

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation / Evidence
575	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.21%	CORRECT	STRICT		1.00	The subject, predicate, and object match the DB candidate exactly (STRICT match for object after normalizing percentage symbol). Evidence: subject: 575, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 12.21
580	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	CORRECT	STRICT		1.00	The subject, predicate, and object match the DB candidate exactly (STRICT match for object after normalizing percentage symbol). Evidence: subject: 580, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 11.91
579	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	~12.06%	FABRICATED_FACT	NONE	MULTIPLE	0.00	The predicate is known, but neither the subject nor the object appear in any triple using this predicate, and the object makes a concrete verifiable claim.
835	DOWN_PAYMENT_REQUIRED_A_S_UPFRONT_BORROWER_CASH_CONTRIBUTION	~2.8% or lower	PREDICATE_MISMATCH	NONE	PREDICATE	0.60	The subject exists, but the predicate is not a valid relation for it in the DB candidates. Evidence: subject: 835, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 835, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 4.91; subject: 835, predicate: SCORE_LADDER_ADJACENT_TO_NEXT_HIGHER_FIVE_POINT_BAND, object: 840

Triple Audit Trail (Original → Canonical → Verdict)

Full pipeline trace for every triple. UNVERIFIABLE* means the triple had no verdict (e.g. dropped due to output truncation).

Orig. Subject	Orig. Predicate	Orig. Object	Confidence	Canon. Predicate	Canon. Object	Verdict
575	HAS_INTEREST_RATE	12.21%	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.21%	CORRECT
580	HAS_INTEREST_RATE	11.91%	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	CORRECT
579	HAS_ESTIMATED_INTEREST_RATE	~12.06% (estimated)	MEDIUM	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	~12.06%	FABRICATED_FACT
835	HAS_ESTIMATED_DOWN_PAYMENT	~2.8% or lower	MEDIUM	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	~2.8% or lower	PREDICATE_MISMATCH

Verdict Legend

Verdict	Description
CORRECT	All three components (subject, predicate, object) verified against DB.
RULE_VIOLATION	One or more components contradict DB evidence.
MULTI_RULE_VIOLATION	Violation spans multiple DB triples simultaneously.
UNVERIFIABLE	No usable DB evidence found for any component.
PREDICATE_MISMATCH	Subject real, but stated predicate is not a valid relation for it.
FABRICATED_FACT	Triple asserts a relationship that does not exist anywhere in DB.